

Runner	HPC	Container	Status	Time (s)
cwltool	LUMI	none	1	87.0
cwltool	LUMI	singularity	IN	79.0
cwltool	MareNostrum 5	none	OK	8.0
cwltool	MareNostrum 5	singularity	1	3.0
cwltool	Laptop	none	OK	3.0
cwltool	Laptop	docker	OK	5.0
cwltool	Laptop	singularity	IN	4.0
streamflow	LUMI	none	1	113.0
streamflow	LUMI	singularity	TO	300.0
streamflow	MareNostrum 5	none	1	16.0
streamflow	MareNostrum 5	singularity	1	1.0
streamflow	Laptop	none	OK	2.0
streamflow	Laptop	docker	OK	3.0
streamflow	Laptop	singularity	TO	31.0
toil	LUMI	none	1	124.0
toil	LUMI	singularity	IN	113.0
toil	MareNostrum 5	none	1	18.0
toil	MareNostrum 5	singularity	1	10.0
toil	Laptop	none	OK	11.0
toil	Laptop	docker	OK	18.0
toil	Laptop	singularity	IN	14.0

Table 1: Simple MPI workflow execution results across runners, platforms, and container backends with baseCommand: mpirun (TO: time-out, IN: invalid)

Runner	HPC	Container	Status	Time (s)
cwltool	LUMI	none	OK	4.0
cwltool	LUMI	singularity	1	5.0
cwltool	MareNostrum 5	none	OK	3.0
cwltool	MareNostrum 5	singularity	1	2.0
cwltool	Laptop	none	OK	3.0
cwltool	Laptop	docker	OK	4.0
cwltool	Laptop	singularity	1	3.0
streamflow	LUMI	none	1	3.0
streamflow	LUMI	singularity	1	3.0
streamflow	MareNostrum 5	none	1	1.0
streamflow	MareNostrum 5	singularity	1	1.0
streamflow	Laptop	none	1	2.0
streamflow	Laptop	docker	1	1.0
streamflow	Laptop	singularity	1	1.0
toil	LUMI	none	OK	14.0
toil	LUMI	singularity	1	20.0
toil	MareNostrum 5	none	1	8.0
toil	MareNostrum 5	singularity	1	9.0
toil	Laptop	none	OK	11.0
toil	Laptop	docker	1	4.0
toil	Laptop	singularity	1	14.0

Table 2: Simple MPI workflow execution results across runners, platforms, and container backends with MPIRequirement (TO: time-out, IN: invalid)